

Mathematical Foundations of Political Methodology

Lecture 4

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Derivative: Motivation

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 - Intuition for the derivative
 - Important tricks for taking derivatives
 - Mean Value Theorem

Line and Slope

A line can be represented by the equation

$$y = a * x + b$$

a is the slope and b the y -intercept

Line and Slope

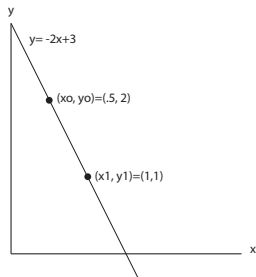
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Rise over Run: For any points on the line, (x_o, y_o) and (x_1, y_1) , the slope

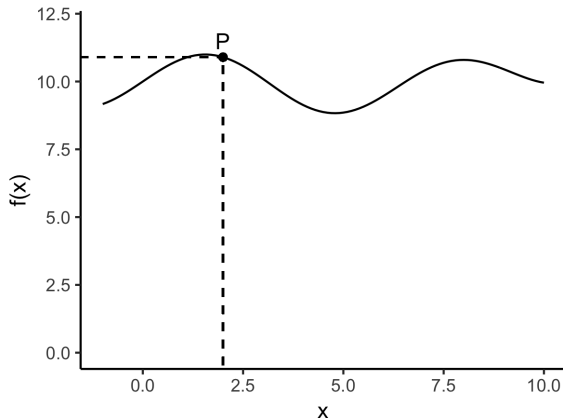
$$a = \frac{y_1 - y_o}{x_1 - x_o}$$



$$\begin{aligned} a &= (y_1 - y_o) / (x_1 - x_o) \\ &= (1 - 2) / (1 - 5) \\ &= -2 \end{aligned}$$

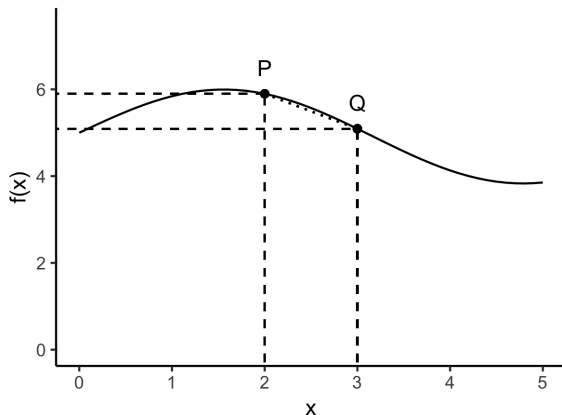
Slope of a Curve

- Much of the time our functions are not straight lines. In these cases, we add the notion of limits to our definition.
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$$\frac{\text{rise}}{\text{run}} = \frac{y_1 - y_0}{x_1 - x_0} = \frac{f(x_0 + h) - f(x_0)}{x_0 + h - x_0} = \frac{f(x_0 + h) - f(x_0)}{h}$$

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As $h \rightarrow 0$, $Q \rightarrow P$ and we have the *derivative* of $f(x)$ at x_0 .

Derivative: Formal Definition

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$$R(x) = \frac{f(x) - f(x_0)}{x - x_0}$$

- $R(x)$ defines the rate of change.
- A derivative will examine what happens with a small perturbation at x_0

Derivative Definition

Definition

Let $f : \mathbb{R} \rightarrow \mathbb{R}$. If the limit

$$\begin{aligned}\lim_{x \rightarrow x_0} R(x) &= \frac{f(x) - f(x_0)}{x - x_0} \\ &= f'(x_0)\end{aligned}$$

exists then we say that f is *differentiable* at x_0 .

If $f'(x_0)$ exists for all $x \in \text{Domain}$, then we say that f is differentiable.

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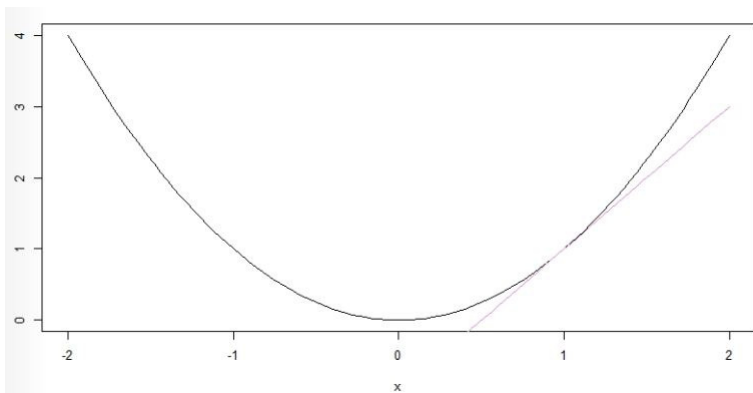
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The figure looks like:

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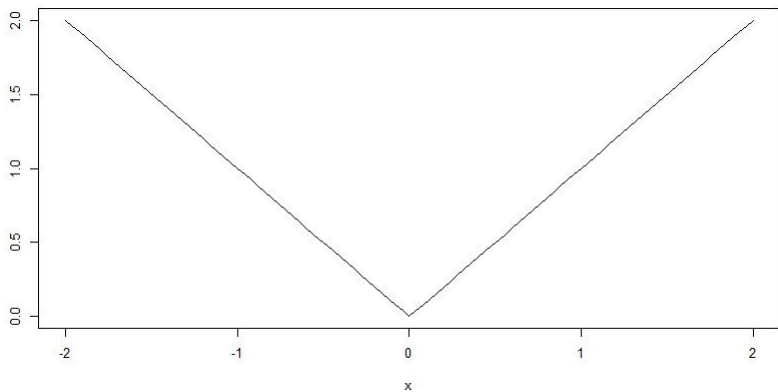
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$$\lim_{x \rightarrow 0} R(x) = \lim_{x \rightarrow 0} \frac{|x|}{x}$$

$\lim_{x \rightarrow 0^-} R(x) = -1$, but $\lim_{x \rightarrow 0^+} R(x) = 1$.

Given this, we say, $|x|$ not differentiable at 0.

Graph of $|x|$



- Derivative
- Derivative and Continuity
- Derivative Rules
- Mean Value Theorem
- Second Derivatives

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Our proof strategy:

- We need $\lim_{x \rightarrow x_0} f(x) = f(x_0)$ (recall last lecture).
- Use direct proof: Start with $f(x)$ that is differentiable.
- Manipulate till we get the formula for the rate of change
$$R(x) = \frac{f(x) - f(x_0)}{(x - x_0)}.$$
- Take the limit, use limit properties, get $f(x_0)$.

Proof of Derivative and Continuity

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Proof of Derivative and Continuity

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$$\lim_{x \rightarrow x_0} f(x) = \lim_{x \rightarrow x_0} [R(x)(x - x_0) + f(x_0)]$$

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$$\begin{aligned}\lim_{x \rightarrow x_0} f(x) &= \lim_{x \rightarrow x_0} [R(x)(x - x_0) + f(x_0)] \\&= \left(\lim_{x \rightarrow x_0} R(x) \right) \left(\lim_{x \rightarrow x_0} (x - x_0) \right) + \lim_{x \rightarrow x_0} f(x_0)\end{aligned}$$

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- **Derivative Rules**
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$$f(x) = \log(x) \quad ; \quad f'(x) = \frac{1}{x}$$

Proof the $\exp(x) \equiv e^x$ is its own derivative

Recall $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$

$$\exp'(x) = \lim_{h \rightarrow 0} \frac{\exp(x+h) - \exp(x)}{h}$$

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Proof the $\exp(x) \equiv e^x$ is its own derivative (cont.)

$$\frac{\exp(h) - 1}{h} = \frac{\sum_{n=0}^{\infty} \frac{h^n}{n!} - 1}{h}$$

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Proof the $\exp(x) \equiv e^x$ is its own derivative (finish)

$$\begin{aligned}\exp'(x) &= \lim_{h \rightarrow 0} \frac{\exp(x+h) - \exp(x)}{h} \\ &= \exp(x) \lim_{h \rightarrow 0} \frac{\exp(h) - 1}{h} \\ &= \exp(x) \lim_{h \rightarrow 0} \left[1 + \frac{h}{2} + \frac{h^2}{3} \dots\right]\end{aligned}$$

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- A person (voter?) has an “ideal point” $p \in \mathbb{R}$
- Person prefers policies that are closer to his/her ideal point: receives utility from policy x

$$u(x) = -(x - p)^2$$

$$u'(x) = 2p - 2x$$

Algebra of Derivatives

Theorem

*Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ and both are differentiable at $x_0 \in \mathbb{R}$.
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$$h'(x_0) = f'(x_0) + g'(x_0)$$

ii) $h(x) = f(x)g(x)$ is differentiable at x_0 and

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Common to have functions in functions

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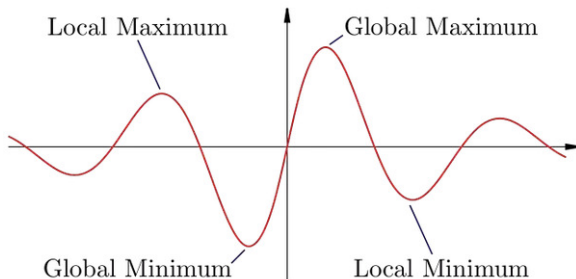
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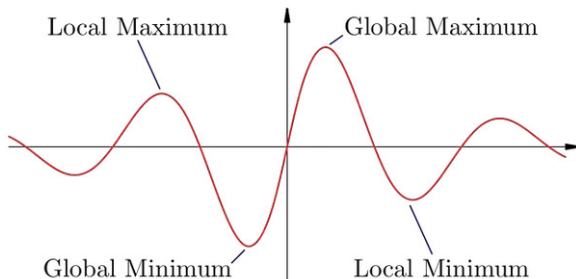
- Derivative
- Derivative Rules
- Mean Value Theorem
- Second Derivatives

Relative Maxima, Minima and Derivatives



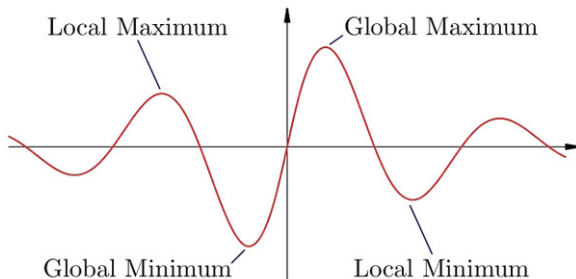
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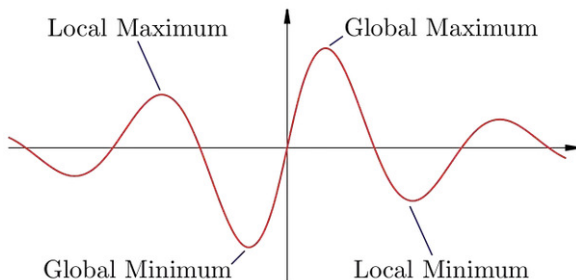
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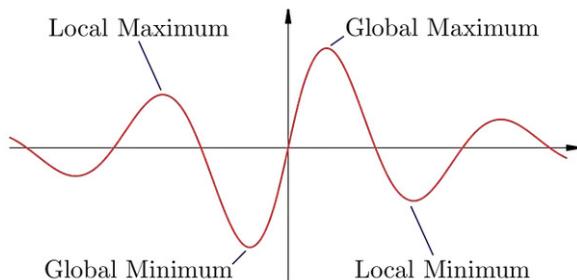
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- We use derivatives to find the maxima and minima.
- We need maxima (or minima) for statistical inference (what is **most** likely), modeling choices (what is **most preferred**), and for modeling systems (what is **stable**).
- The connection between maxima and derivatives is shown with the Mean Value Theorem (MVT).
- To get there we prove a special case of MVT called “Rolle’s Theorem.”

Aside: Naming Conventions of Theorems

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- Stigler's Law of Eponymy.
 - “No scientific discovery is named for the person who discovered it.”
- Rolle's Theorem was proved by Cauchy (1823), but first stated by a 12th century Indian mathematician Bhaskara II

Derivatives and local maxima/minima

Theorem

Rolle's Theorem Suppose $f : [a, b] \rightarrow \mathbb{R}$ and f is continuous on $[a, b]$ and differentiable on (a, b) . Then if $f(a) = f(b) = 0$, there is $c \in (a, b)$ such that $f'(c) = 0$.

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 - Show that the left hand limit can't be positive, and the right hand limit can't be negative.
 - Then all that is left is 0.

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In order for these two limits to be equal,

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It must be that: $f'(c) = 0$.

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- The contrapositive of Rolle's Theorem states that if there is not a $c \in (a, b)$ such that $f'(c) = 0$, then $f(a) \neq f(b)$.
- But $f(d) = 1 = f(e)$, which is a contradiction.
- There must be only one solution.

Mean Value Theorem

Theorem

If $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b) , then there is a $c \in (a, b)$ such that

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- That is, if there is a line connecting two points of a function, the slope will be equal to the derivative somewhere between the two points.
- The proof strategy is to set up for an application of Rolle's theorem.

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So $g(x)$ is like $f(x)$, in that if you evaluate it at a we get $f(a)$, but if you evaluate it at b you still get $f(a)$. This means it starts and ends at the same place, so it must turn around.

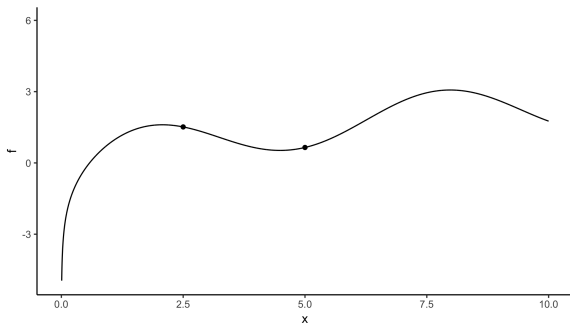
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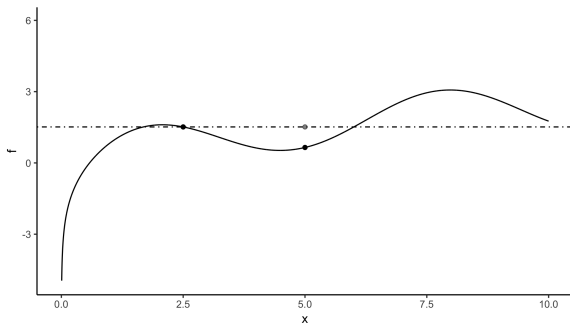
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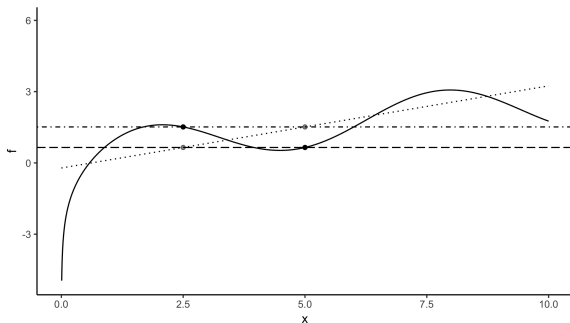
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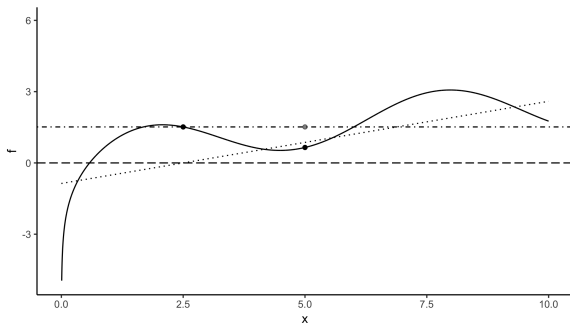
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Intuition for Proof Strategy

$$g(x) = f(x) - \frac{(f(b) - f(a))}{b - a}(x - a)$$

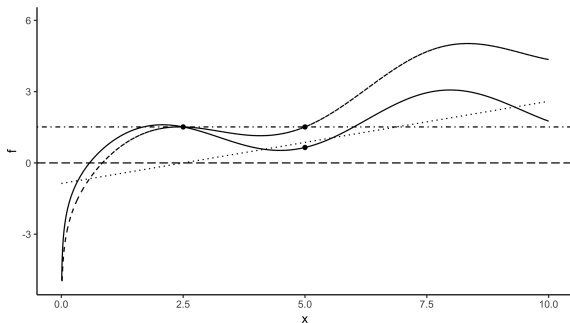
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Applications of Mean Value Theorem

- Speeding tickets: If the police can measure how long it takes me to get from point A to point B, at some point I was driving that speed.
- Used to derive popular approximations to functions (including Taylor expansion)
- We will use it in the proof of the Fundamental Theorem of Calculus relating derivatives to integrals.

- Derivative
- Derivative Rules
- Mean Value Theorem
- Second Derivatives

Second Derivatives

If the derivative is a slope, the second derivative is the slope of a slope.

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Definition

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable. Recall we write this as f' and suppose that $f' : \mathbb{R} \rightarrow \mathbb{R}$. Then if the limit,

$$\lim_{x \rightarrow x_0} R(x) = \frac{f'(x) - f'(x_0)}{x - x_0}$$

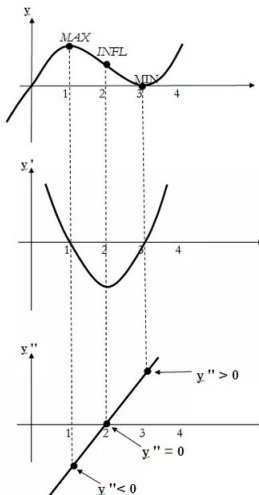
*exists, we call this the **second derivative** at x_0 , $f''(x_0)$.*

Visualization of First and Second Derivative

$$\begin{aligned}y &= x(x-3)^2 \\&= x^3 - 6x^2 + 9x \\&\text{(cubic curve)}\end{aligned}$$

$$\begin{aligned}y' &= 3x^2 - 12x + 9 \\&= 3(x^2 - 4x + 3) \\&= 3(x-1)(x-3) \\&\text{(parabola)}\end{aligned}$$

$$\begin{aligned}y'' &= 6x - 12 \\&\text{(line graph)}\end{aligned}$$



Examples of Second Derivatives

$$f(x) = x$$

$$f'(x) = 1$$

$$f''(x) = 0$$

Examples of Second Derivatives

$$\begin{aligned}f(x) &= e^x \\f'(x) &= e^x \\f''(x) &= e^x\end{aligned}$$

Examples of Second Derivatives

$$\begin{aligned}f(x) &= \log(x) \\f'(x) &= \frac{1}{x} = x^{-1} \\f''(x) &= -\frac{1}{x^2}\end{aligned}$$

Examples of Second Derivatives

$$f(x) = \frac{1}{x}$$

$$f'(x) = \frac{-1}{x^2}$$

$$f''(x) = \frac{2}{x^3}$$

Examples of Second Derivatives

$$\begin{aligned}f(x) &= -x^2 + 20 \\f'(x) &= -2x \\f''(x) &= -2\end{aligned}$$

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Going forward

- Your job is to memorize the product, quotient and chain rules of derivatives.
- We will see how the second derivative is used to evaluate points as relative minima and maxima.